AUC Rideshare App

Milestone I

Mai Abd El-Naby
Jana Elfeky
Kareem Sayed
Mariam Mohamed
Shahd Abou Eldahab
Farida Hazem



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1. Overview of the SW system

What is it about?

- Facilitate carpooling within the AUC community.
- Connect AUC students, faculty, and staff with shared transportation needs.
- Reduce transportation costs and promote eco-friendly commuting.
- Schedule rides in advance or request them in real-time.
- Enhance safety through in-app features like driver preferences.
- Enable seamless communication and payment options between riders and drivers.



1. Overview of the SW system

Users

- AUC Students: Primary users who commute to/from campus.
- AUC Faculty and Staff: Secondary users benefiting from shared rides.
- Drivers and Admins: Offering rides or managing app operations.



Motivation

- Build a stronger sense of community among AUCians.
- Provide safer and more reliable transportation alternatives.
- Reduce the number of single-passenger cars to minimize congestion and environmental impact.

2. The product Backlog

2. The product Backlog

I. Functional requirements

- Users should register with their AUC ID and phone numbers.
- Drivers should register their car information like its model, make, colour, and license plate along with their AUC ID and phone numbers.
- Users should choose preferences during registration (e.g., driver gender, music taste, hobbies, topics to discuss, etc.).
- Users should be able to input their class schedules during registration.
- Users should be able to schedule rides in advance and in real-time.
- Riders and drivers should see each other’s contact info and location when a ride is confirmed to facilitate communication.

2. The product Backlog

I. Functional requirements

- There will be penalties for:
  - Drivers who are late to pick up riders.
  - Riders who are late to meet drivers.
- Users should be able to “friend” other users with similar schedules or preferences to have a go-to ride partner.
- Riders shall be able to pay using cash, Telda, or Instapay.
- There should be a ride review/ rating at the end of each ride which is then recorded on the rider and driver’s profiles
- Before confirming their rides, riders shall be able to include up to three stops.
- The driver shall receive real-time routing updates with each individual stop, if more than one stop is set.

2. The product Backlog

I. Functional requirements

- Users shall be able to report unsafe behavior in the app.
  - Reports shall be sent to admins for review, and users who receive multiple reports will be flagged.
  - Admins shall be able to suspend users who violate safety rules.
  - The system shall store all ride data automatically such as location and driver’s information
- Admins shall be able to review flagged users.
- Notifications shall be sent to users for ride status updates and payment confirmations.

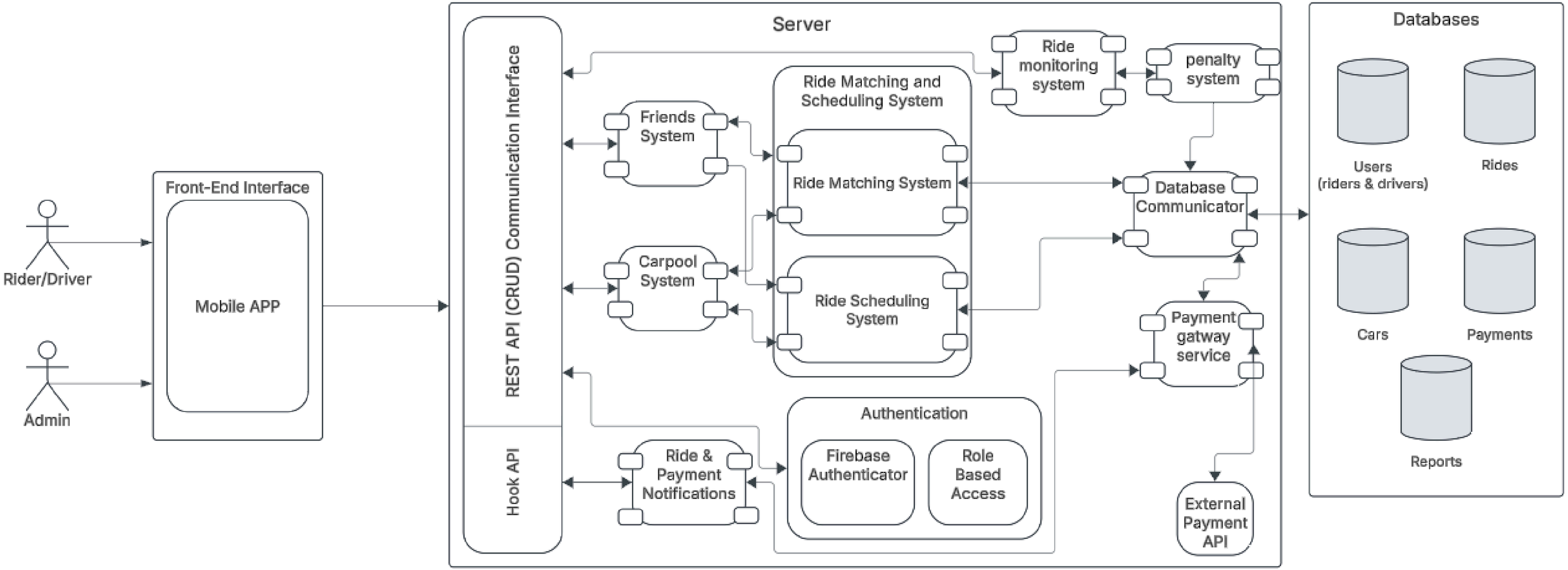
2. The product Backlog

II. Non-Functional requirements

- Authentication should be done using AUC’s database.
- Ride request should time out after searching for a driver for 5 minutes without a response.
- Riders cannot book a ride more than 3 days in advance.
- Location sharing shall be real-time and accurate through a third party API.
- Payment confirmation shall be required for each transaction.
- System should keep track of every ride, with its associated payment and report if any.
- Reports shall be reviewed within 48 hours.
- Servers should have a minimum of 95% uptime.
- Notifications will be in banner style.

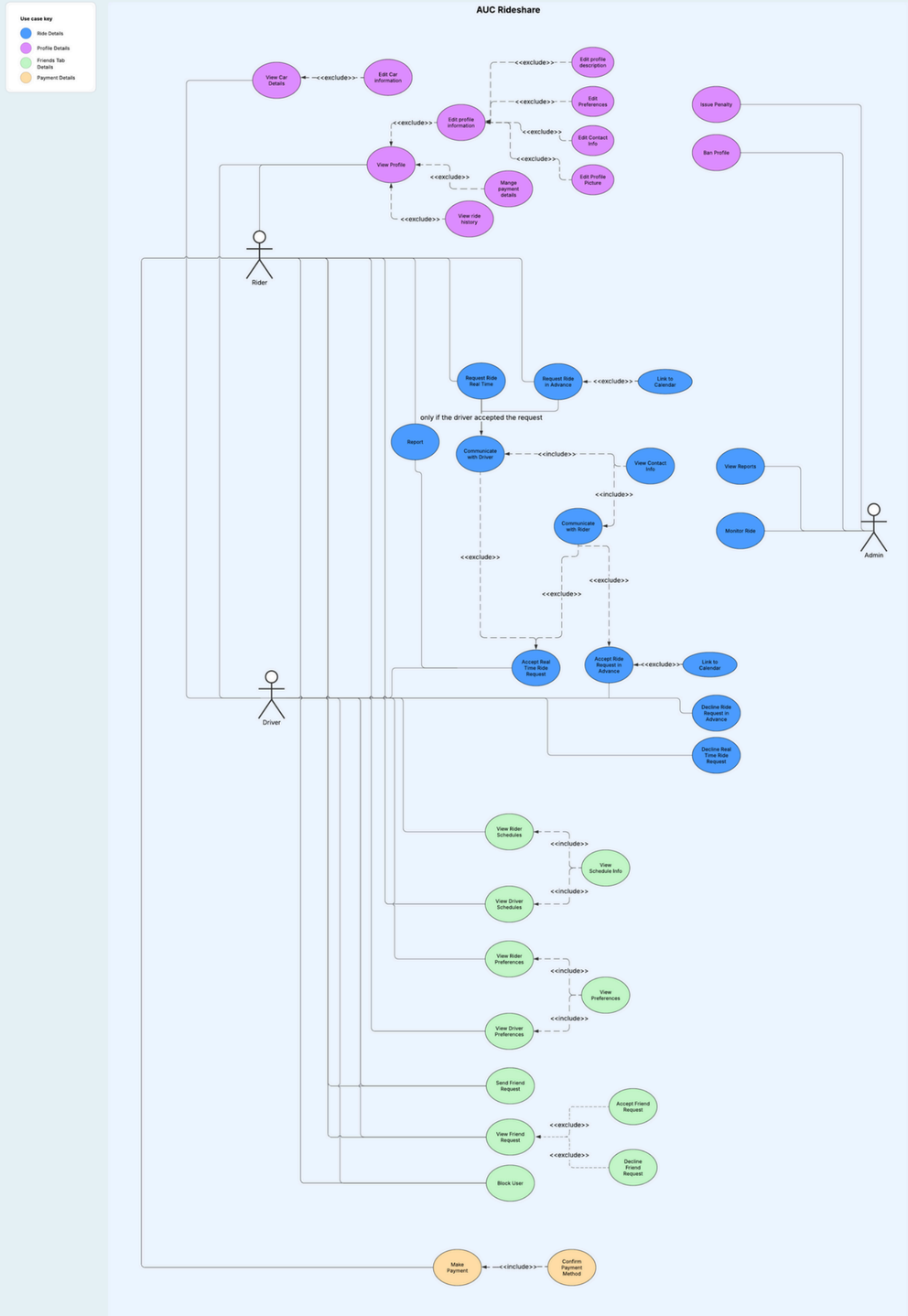
3. High-level Architecture

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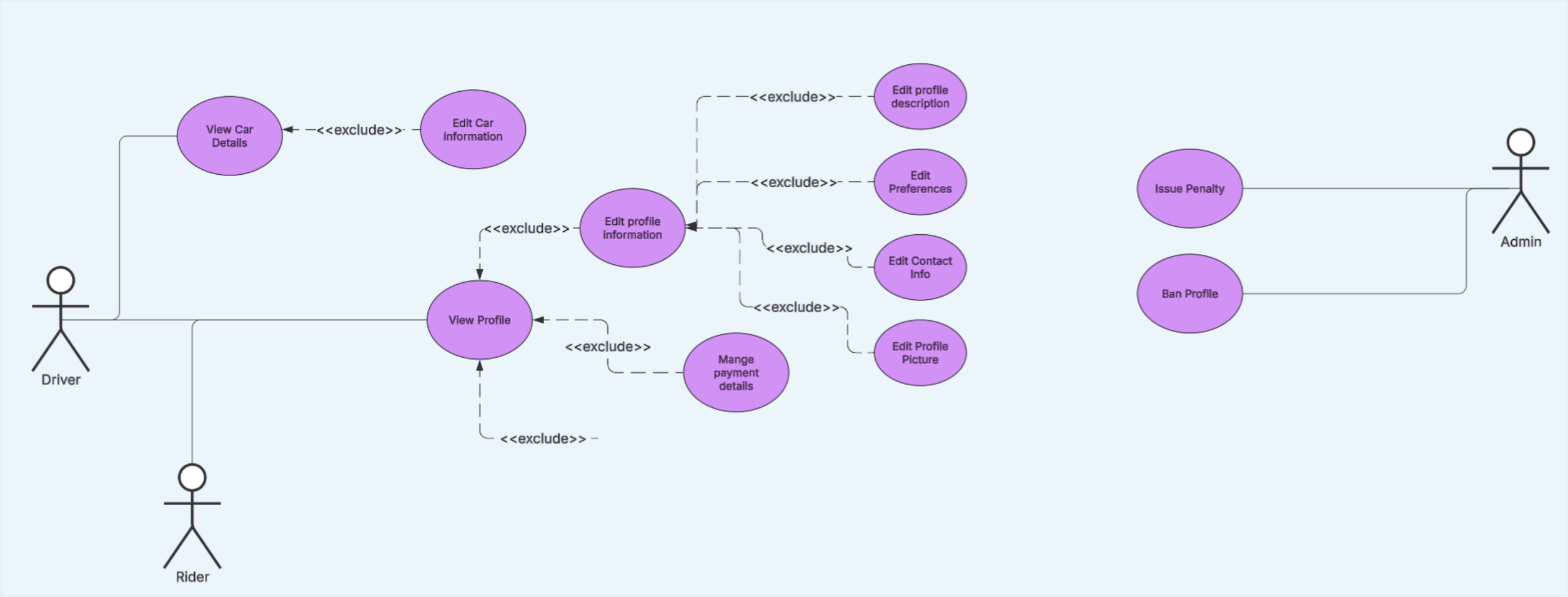


4. Use Case Diagram

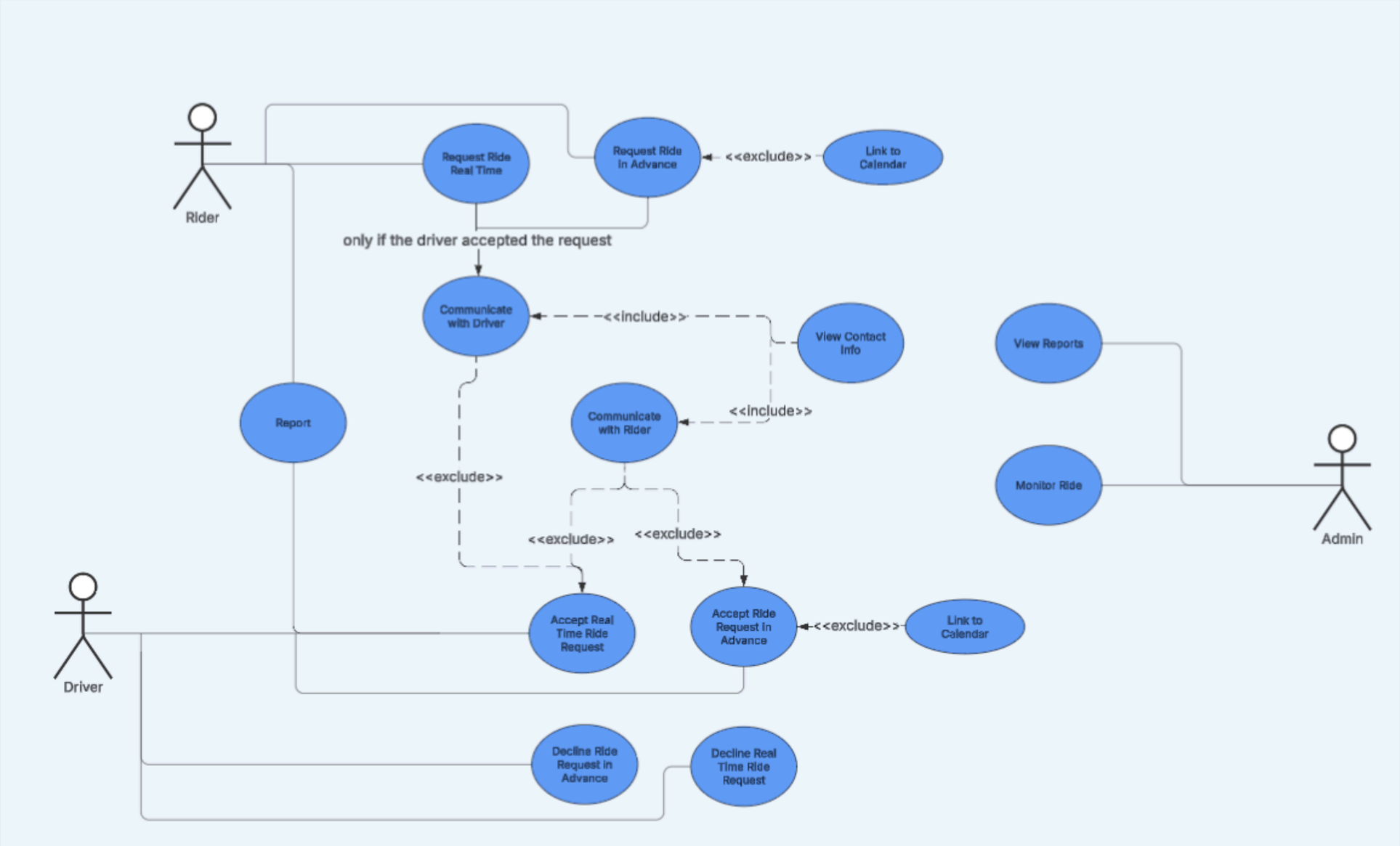
Use Case Diagram



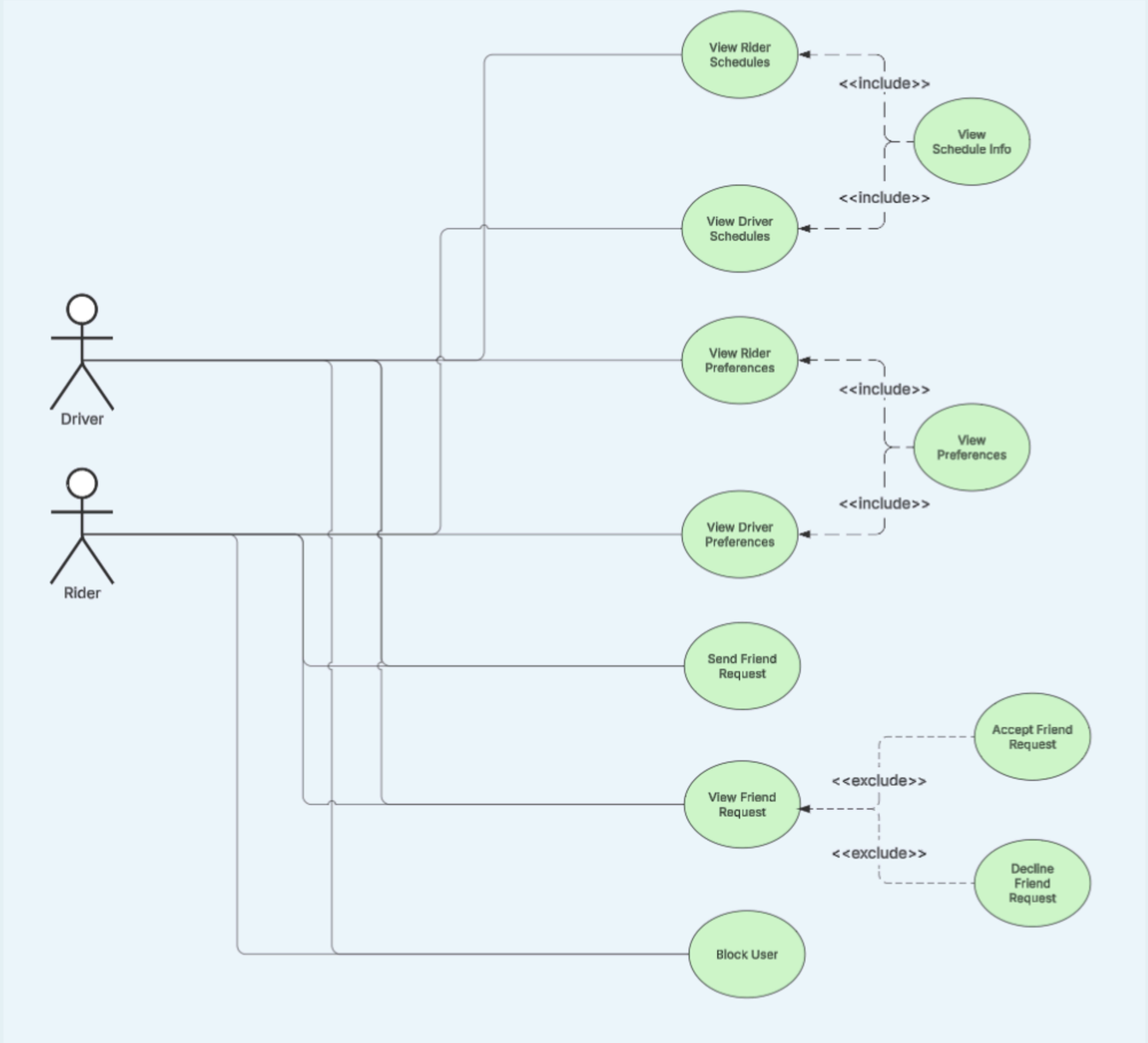
Profiles


View Car Details
<<exclude>>
Edit Car information
<<exclude>>
Edit profile description
<<exclude>>
Edit Preferences
<<exclude>>
Edit profile information
<<exclude>>
Edit Contact Info
<<exclude>>
Edit Profile Picture
Driver
View Profile
<<exclude>>
Mange payment details
<<exclude>>
Rider
Issue Penalty
Ban Profile
Admin

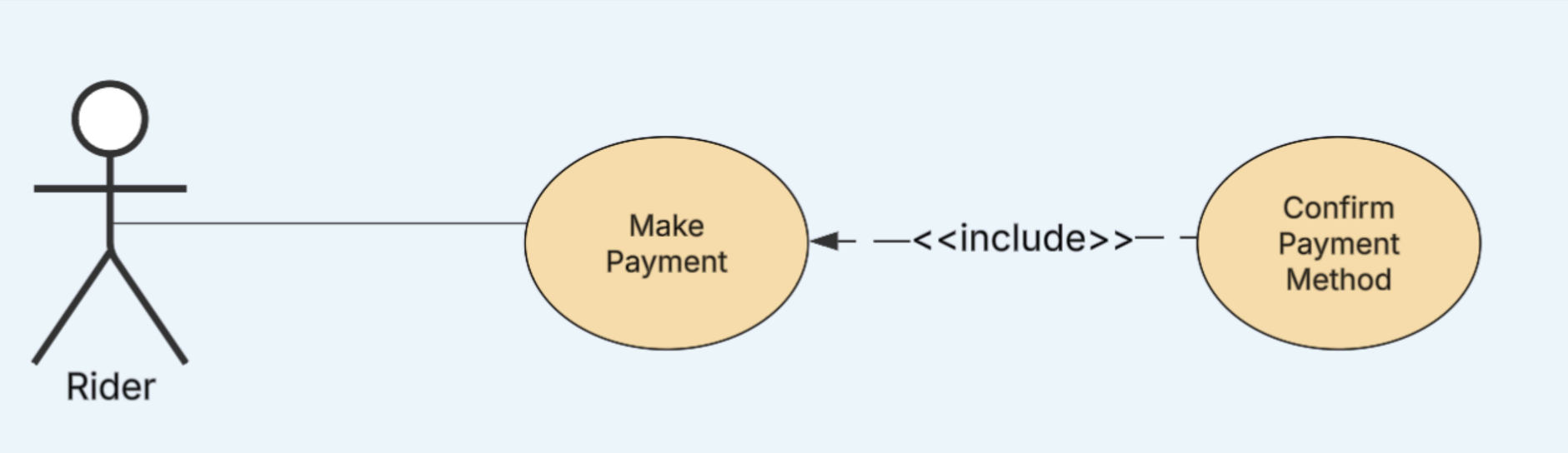
Ride Details



Friends Tab


View Rider Schedules
<<include>>
View Schedule Info
<<include>>
View Driver Schedules
View Rider Preferences
<<include>>
View Preferences
<<include>>
View Driver Preferences
Driver
Rider
Send Friend Request
Accept Friend Request
<<exclude>>
View Friend Request
<<exclude>>
Decline Friend Request
Block User

Payment



5. Identified risks

AUC Cooperation

The app requires integration with AUC's database for authentication.

User Adoption

Convincing students to adopt the app and use it regularly may be difficult.

Safety Concerns

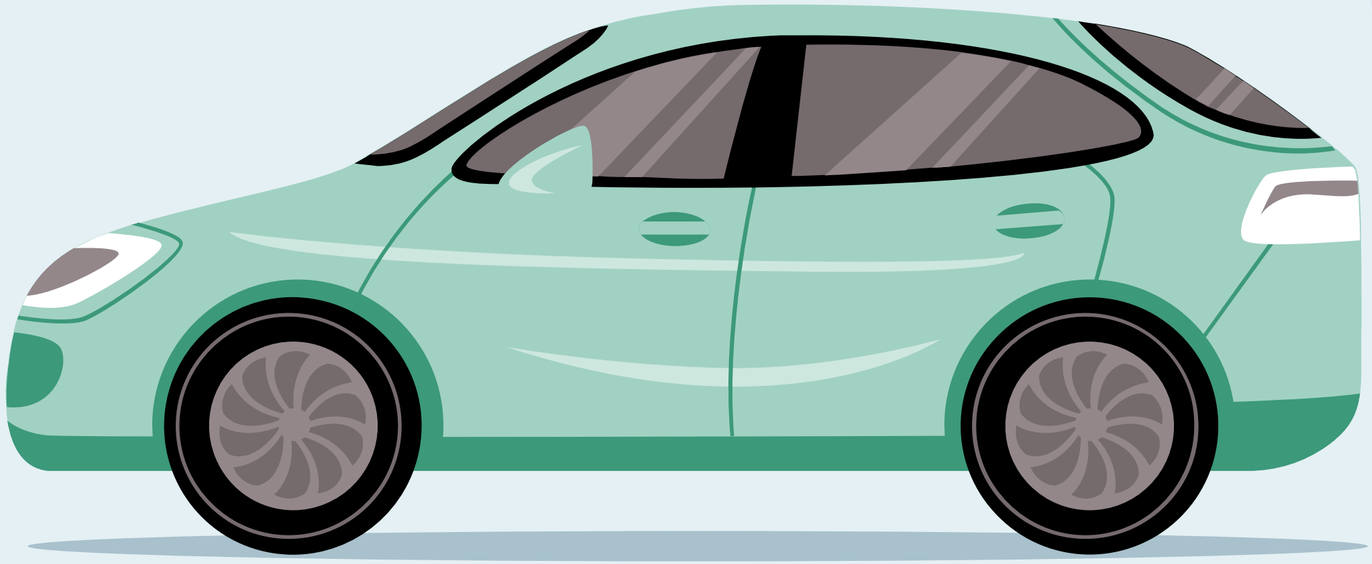
Maximizing the safety of riders and drivers, especially when sharing personal information like phone numbers and locations. (Will depend on the third party API we will use)

Payment & Penalties

- Ensuring secure transactions.
- Enforcing penalties for late arrivals could lead to disputes between users or users refusing to use the app.

6. Semester Backlog

- Implement all user profile interfaces
- Implement safety and reporting system
- Develop the friends feature and payment methods

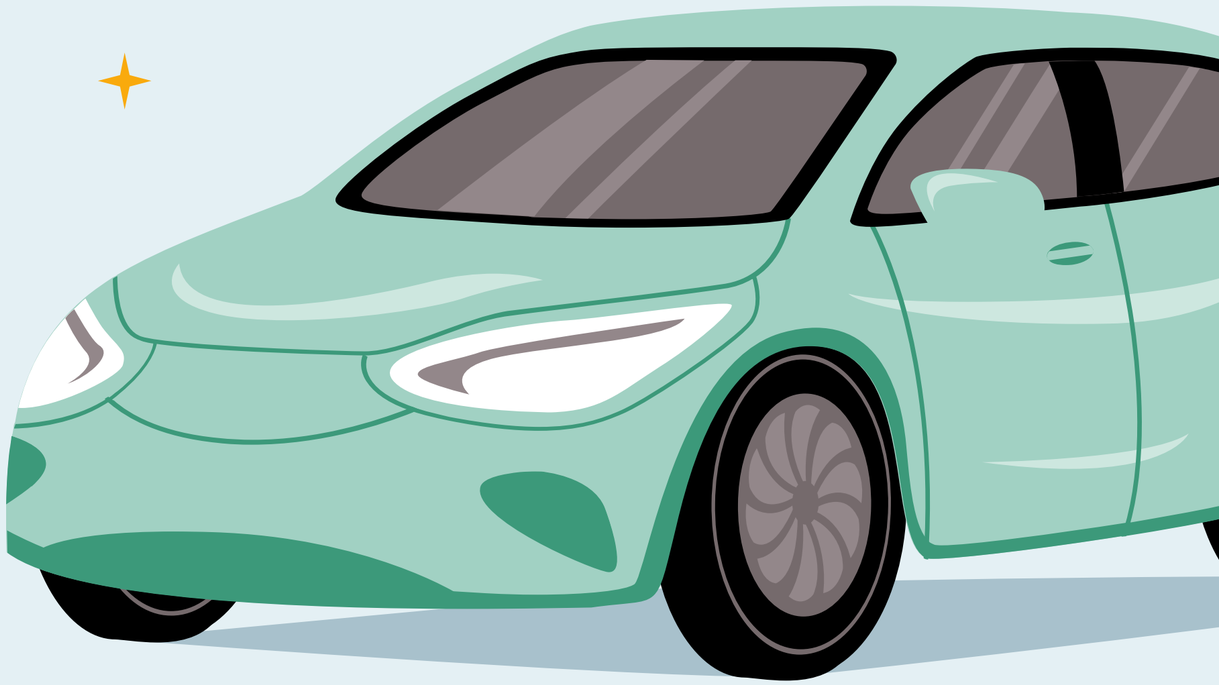


7. Conclusion

- Identification of various functional and non functional requiremnts
- Desgined the high level architecture for the feature that will be developed
- Desgined the Use Case digram to portray the various action taken by various users such as the rider and the driver
- identified the various risks

THANKS!

DO YOU HAVE ANY QUESTIONS?



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